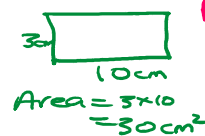
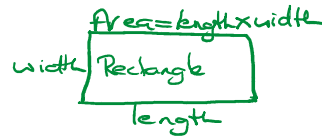


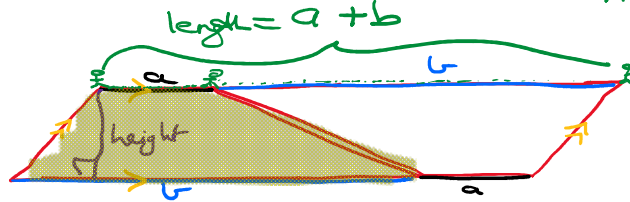
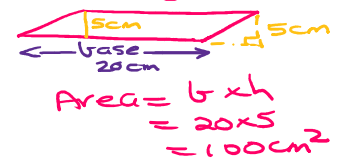
Questions - Tutorial

Surface Area - which height

Trapezium
Trapezoid



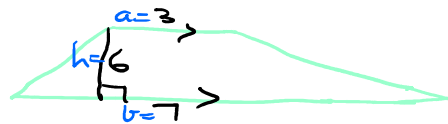
Area = base $\times$ height



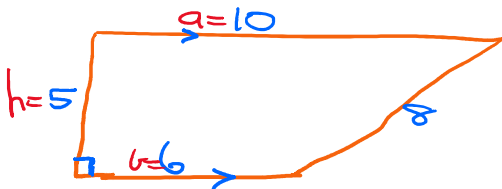
Parallelogram Area = base $\times$ height

$$\text{Trapezoid Area} = \frac{1}{2}(a+b) \times \text{height} = \frac{1}{2}(a+b) \times h$$

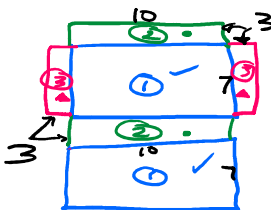
a & b are parallel sides.



$$\begin{aligned} \text{Area} &= \frac{1}{2}(a+b) \times h \\ &= \frac{1}{2}(3+7) \times 6 \\ &= \frac{1}{2} \times 10 \times 6 \\ &= 5 \times 6 \\ &= 30 \text{ cm}^2 \end{aligned}$$

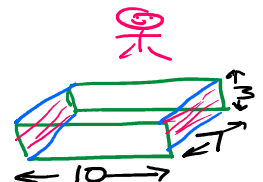


$$\begin{aligned} \text{Area} &= \frac{1}{2}(a+b) \times h \\ &= \frac{1}{2}(10+6) \times 5 \\ &= \frac{1}{2} \times 16 \times 5 \\ &= 8 \times 5 \\ &= 40 \text{ cm}^2 \end{aligned}$$



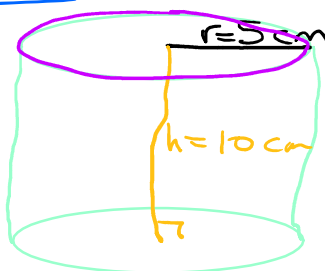
The surface area of a net
Front + back + left + right + top + bottom

$$\begin{aligned} \text{S.A.} &= 2 \times (1) + 2 \times (2) + 2 \times (3) \\ &= 2 \times 10 \times 7 + 2 \times 10 \times 3 + 2 \times 7 \times 3 \\ &= 140 + 60 + 42 \\ &= 242 \text{ cm}^2 \end{aligned}$$



$$\begin{aligned} \text{Volume} &= l \times w \times h \\ &= 10 \times 7 \times 3 \\ &= 210 \text{ cm}^3 \end{aligned}$$

Cylinder



Surface Area =

$$\begin{aligned} \text{Area} &= \pi r^2 \\ &= \pi \times (5)^2 = 25\pi \\ &= 78.54 \text{ cm}^2 \end{aligned}$$

$$\text{Area} = \frac{2\pi r}{h}$$

$$\begin{aligned} A_{\square} &= 2 \times \pi \times 5 \times 10 \quad (100\pi) \\ &= 314.2 \end{aligned}$$

$$\text{circumference} = \pi \times D = 2 \times \pi \times r$$

$$\begin{aligned} \text{Total surface} &= 157.08 + 314.2 \\ &= 471.28 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned}
 \text{Volume} &= \text{Area} \times \text{height} \\
 &= A_0 \times h \\
 &= \pi r^2 h \\
 &= \pi (5)^2 \times 10 \\
 &= 250\pi = 785.4 \text{ cm}^3
 \end{aligned}$$

measuring in metres (meters)

$$\frac{\text{Volume}}{\text{m}^3}$$

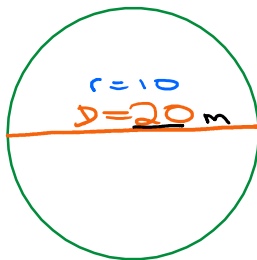
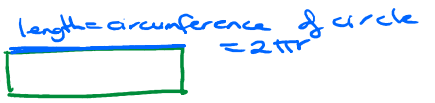
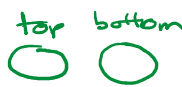
$$\frac{\text{Area (surface)}}{\text{m}^2} \quad \frac{\text{Perimeter/length (circumference)}}{\text{m}}$$

$$A = \pi r^2 \quad \text{circle} \quad \text{cylinder}$$

$$V = \pi r^2 h$$

$$* \text{S.A. cylinder} = 2\pi r^2 + 2\pi r h$$

$$C = \pi D \quad \text{circumference } C = 2\pi r$$



$$\text{circumference} = \pi \times D = 3.142 \times 20 = 62.84 \text{ m}$$

$$\begin{aligned}
 \text{Area} &= \pi r^2 \\
 &= \pi \times (10)^2 \\
 &= \pi \times 100 \quad (10^2 = 100) \\
 &= 314.2
 \end{aligned}$$

$$\text{Surface Area (SA)} =$$

$$\begin{aligned}
 &= 2\pi r^2 + 2\pi r h \\
 &= 2\pi (10)^2 + 2\pi (10)(50) \\
 &= 2 \times \pi \times 100 + 1000 \times \pi \\
 &= 200\pi + 3142 \\
 &= 628 + 3142 \\
 &= 3770 \text{ m}^2 / 3768
 \end{aligned}$$

$$\begin{aligned}
 \text{A semicircle} &= \frac{\pi r^2}{2} \\
 &= \frac{314}{2} = 157.
 \end{aligned}$$

$$\begin{aligned}
 \text{Perimeter of semicircle} &= \frac{\pi D}{2} + \frac{10 \times 10}{2} \\
 &= \frac{62.84}{2} + 20 \\
 &= 31.42 + 20 \\
 &= 51.4 \text{ m}
 \end{aligned}$$

$$\begin{array}{r}
 31.42 \\
 + 20 \\
 \hline
 51.42
 \end{array}$$



$$A = \pi r^2 \quad r=10$$



$$\text{Volume} = \pi r^2 h$$

$$\begin{aligned}
 &= \pi (10)^2 \times 50 \\
 &= \pi \times 100 \times 50 \\
 &= 15708 \text{ m}^3 \quad (5000\pi)
 \end{aligned}$$

pyramid

meets at a point



prism each slice is the same

each face is the same //

Triangular prism

triangle height 6



top sides base

Surface area of triangular prism



$$c = \sqrt{2.5^2 + 6^2} = 6.5$$

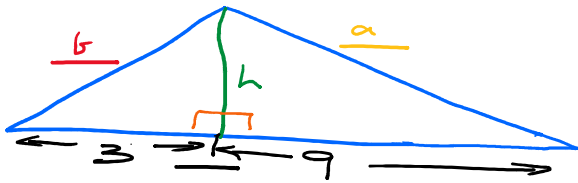
Surface area of triangular prism

2x

$$S_{ft} = \frac{1}{2}bh + 2 \times (6.5 \times 15 + 6.5 \times 15 + 5 \times 15)$$

$$= \frac{1}{2} \times 5 \times 6 + 97.5 + 97.5 + 75$$

$$= 285 \text{ m}^2$$

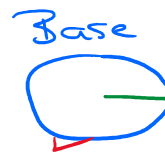
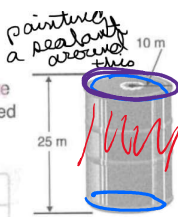


$$h^2 + 9^2 = a^2$$

$$6^2 = h^2 - 3^2$$

4. To prevent leakage, an oil storage tank is coated with a sealant. How many square meters will need to be coated to cover the container?

Solve for: perimeter area



circumference of circle.

$$A = 62.8 \times 25$$

height = 25 m

what shape is the red?

Area of Top & base, $r = 10$ $(10)^2 = 100$

$$A = \pi r^2 + \pi r^2$$

$$= \pi (10)^2 + \pi (10)^2$$

$$= 314 + 314$$

$$= 628 \text{ m}^2$$

$$\pi \times 100 = 3.14 \times 100 = 314$$

$$C = \pi \times D = \pi \times 20 = 62.8$$

$D = 20$ $\frac{2 \times r}{2 \times 10}$ $2\pi r$



$$\text{Area} = L \times V_{25}$$

$$= 62.8 \times 25$$

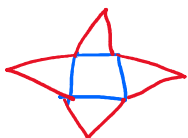
$$= 1570$$

$$= 1570 \text{ m}^2$$

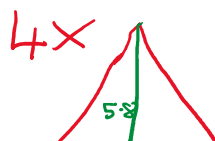
$$= 2198 \text{ m}^2 \text{ (height of 25)}$$

$$\text{Total Area} = 628 + 1570$$

$$= 2198 \text{ m}^2 \text{ (height of 25)}$$

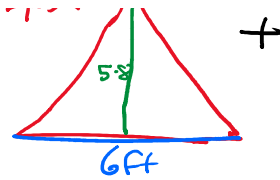
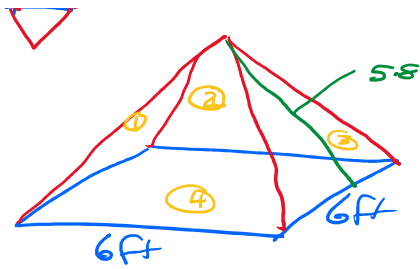


Net $\Rightarrow$ surface area



+

$$\text{Area} = 6 \times 6 = 36 \text{ ft}^2$$



$$\begin{aligned} \text{Area} &= 6 \times 6 \\ &= 36 \text{ ft}^2 \end{aligned}$$

$$\begin{aligned} \text{Area} &= \frac{1}{2} \times \text{base} \times \text{height} \\ &= \frac{1}{2} \times 6 \times 5.8 \\ &= 17.4 \text{ ft}^2 \end{aligned}$$

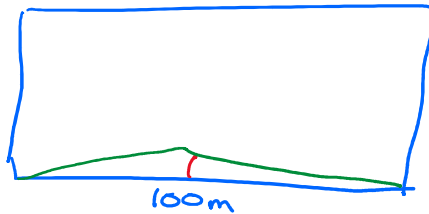


$$4 \times 17.4 + 36$$

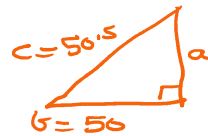
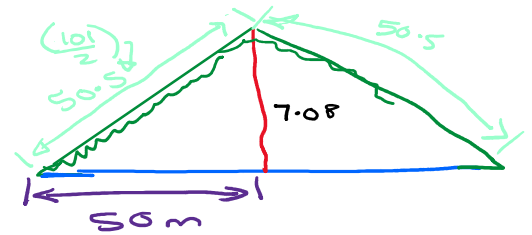
$$\begin{aligned} \text{Total Area} &= 69.6 + 36 \\ &= 105.6 \text{ ft}^2 \end{aligned}$$

how tall is the pole?

- 10m
- 6m
- 15m ✓
- 7m ✓✓
- 1m



bunting 101m
101m



$$\begin{aligned} \frac{a^2}{c^2} &= \frac{b^2}{c^2} \\ c^2 &= a^2 + b^2 \\ a^2 &= c^2 - b^2 \end{aligned}$$

$$\begin{aligned} a^2 &= c^2 - b^2 \\ &= (50.5)^2 - (50)^2 \\ &= 2550.25 - 2500 \\ a^2 &= 50.25 \\ \sqrt{a^2} &= \sqrt{50.25} \\ a &= 7.08 \end{aligned}$$

similar = bigger
congruent = smaller
congruent = the same



similar = same shape but different size.

similar



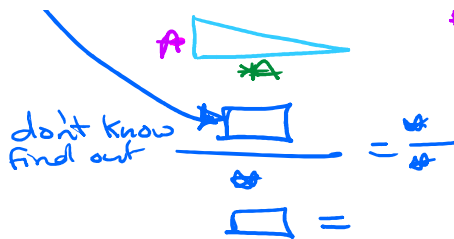
$$3 \div 3 = \frac{3}{3} = 1$$

$$\begin{aligned} 15 &\xrightarrow{\times 1.5} 22.5 \\ 10 &\xrightarrow{\div 1.5} 6.67 \end{aligned}$$



$$\begin{aligned} \frac{A}{B} &= \frac{x}{x} \\ \frac{A}{B} &= \frac{B}{B} \end{aligned}$$

$$\begin{aligned} \frac{x}{3} &= \frac{10}{15} \\ \frac{x}{3} &= \frac{2}{3} \\ x &= 2 \end{aligned}$$



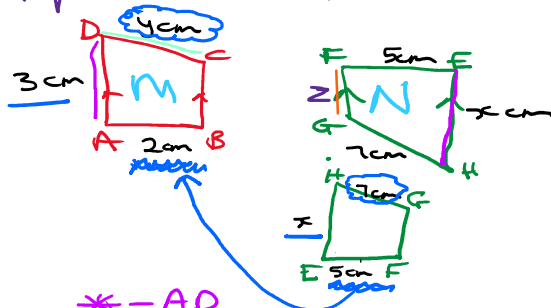
$$\frac{A}{B} = \frac{A}{B}$$

$$\frac{A}{B} = \frac{A}{B}$$

$$= 4.5$$

$$= \frac{9}{2} = 4.5$$

Rhombus = square pushed over
Trapezoid (one pair of // sides) (trapezium).



$$\frac{EH}{AD} = \frac{FG}{AB}$$

$$\frac{EH}{AD} = \frac{FG}{AB}$$

$$* = AD$$

$$* = CB$$

which sides would we use to find the scale factor (make bigger/smaller)
 $\frac{EH}{AD} = \frac{FG}{AB}$
 $\frac{5}{3} = \frac{7}{2}$
 $\frac{5}{3} \times 2 = \frac{7}{2} \times 3$
 $\frac{10}{3} = \frac{21}{2}$
 $\frac{10}{3} \times 3 = \frac{21}{2} \times 3$
 $10 = \frac{63}{2}$
 $10 \times 2 = \frac{63}{2} \times 2$
 $20 = 63$
 this is needed to work out scale factor

$$\frac{EH}{EF} = \frac{AD}{AB}$$

$$\frac{EH}{5} = \frac{3}{2}$$

$$5 \times \frac{EH}{5} = \frac{3}{2} \times 5$$

$$EH = \frac{15}{2} = 7.5$$

$$\frac{DC}{GH} = \frac{AB}{EF}$$

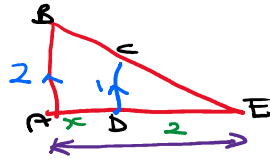
$$\frac{DC}{7} = \frac{2}{5}$$

$$(DC \div 7)$$

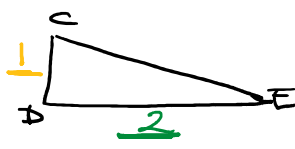
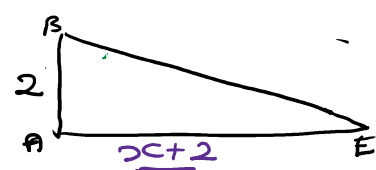
$$7 \times \frac{DC}{7} = \frac{2}{5} \times 7$$

$$DC = 2.8$$

$$\frac{14}{5} = \frac{28}{10} = 2.8$$



Draw the 2 triangles that are similar



$$\frac{AB}{CD} = \frac{AE}{DE}$$

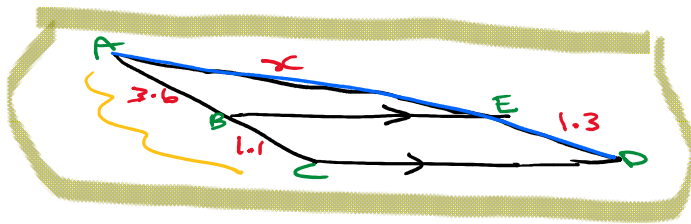
$$2 \times \frac{2}{1} = \frac{x+2}{2}$$

$$4 = x+2$$

$$\frac{AB}{CD} = \frac{PL}{PE} \quad 2x \cdot \frac{1}{1} = \frac{21 \cdot 2}{2} \times 2$$

$$4 = x + 2$$

$$2 = x \quad \checkmark$$



1. Draw 2 triangles
2. Set up ratio for similar triangles.

1. Drawn two separate similar Δ 's
2. Create a ratio of similar sides
3. Solve for unknown value.

$$\frac{4.7}{x+1.3} = \frac{3.6}{x}$$

$$\frac{10}{10} = 1$$

$$\frac{20}{20} = 1$$

$$\frac{x}{x} = 1 \quad \checkmark$$

$$\frac{x+1.3}{x+1.3} = 1$$

$$x+1.3 = x+1.3$$

$$x \times 4.7 = \frac{3.6(x+1.3)}{x}$$

$$4.7x = 3.6(x+1.3)$$

get rid of brackets

$$4.7x = 3.6x + 4.68$$

$$* 4.7x - 3.6x = 4.68$$

$$\frac{1.1x}{1.1} = \frac{4.68}{1.1}$$

$$x = 4.25 \quad (4.68 \div 1.1)$$

Find value of x
Goal $\Rightarrow$ $BC = 4.25$

angles in a triangle add to 180°

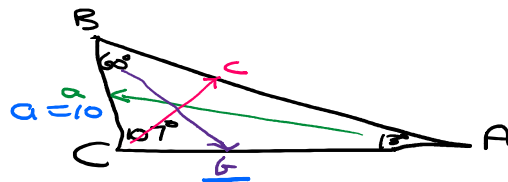
$$A = 13^\circ$$

$$B = 60^\circ$$

$$C = 180 - (60 + 13) = 107^\circ$$

$$a = 10 \text{ cm}$$

$$b = ?$$



cosine Rule

$$c^2 = a^2 + b^2 - 2ab \cos C$$

Sine Rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$A = 10$$

$$A = 13$$

$$B = 60$$



No angle $= 90^\circ$ $\checkmark$

Yes or No?

$$a^2 + b^2 = c^2$$

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$

cosine Rule
sine Rule.

Sine Rule
 $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

$a=10$
 $A=13$
 $B=60$

$\cos \theta =$
 $\tan \theta =$

don't know

$\frac{b}{\sin B} = \frac{a}{\sin A}$

$\frac{b}{\sin 60} = \frac{10}{\sin 13}$

$b = \frac{10 \times \sin(60)}{\sin(13)}$
 $= 38.5$

$\frac{10 \times 0.866}{0.225}$
 $=$

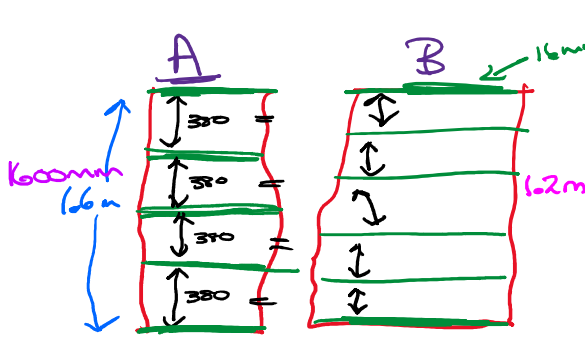
find value of b?

calculator DEG mode not RAD

$\sin(30) = 0.5$ ✓ DEG

$\sin(60) = \frac{\sqrt{3}}{2}$

$\sin(13) = 0.225$
 $\sin(60) = 0.866$



$1.6m = \frac{1600mm}{1000mm}$

$x = \text{height of cupboard} = 1.6m = 1600mm$

$y = \text{each shelf is } 16mm \text{ thick.}$

$z = \text{gap between shelves (equal).}$

What would be the unknown that I want to know?

shelf = 16mm thick.

no of gaps + thicknesses of shelves

$x = 5 \times \text{shelf} + 4 \times \text{gaps}$
 $1600 = 5 \times 16 + 4z$

$1600 = 80 + 4z$

$\frac{1520}{4} = \frac{4z}{4}$

$z = 380mm \text{ for each gap.}$

$4 \overline{) 1520}$

40 tenths

$40 \text{ tenths} = \frac{40}{10}$

height of cupboard

thick shelf

size of gap.

$x = 6y + 5z$

$x = 1.2m = 1200mm$

$1200 = 6 \times 16 + 5z$

$1200 = 96 + 5z$

$1104 = 5z$

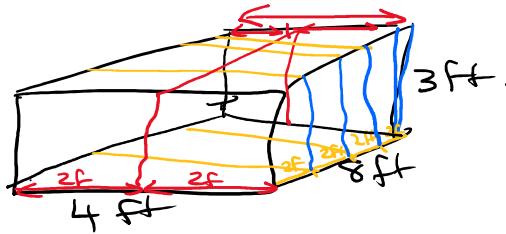
$z = 220.8$

ones tenths

$5 \overline{) 1104.0}$

$5 \times 200mm = 1000mm$

Smaller boxes



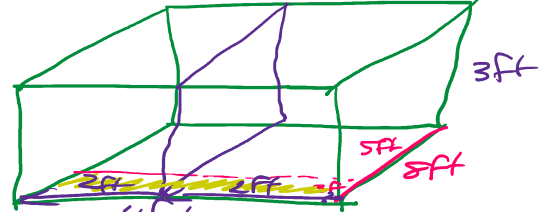
$4 \times 3 \times 8 = 96$

4×24

2×48

$3 \times 2 \times 2 = 12$

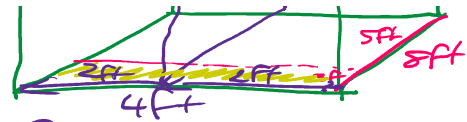
$96 \div 12 = 8$



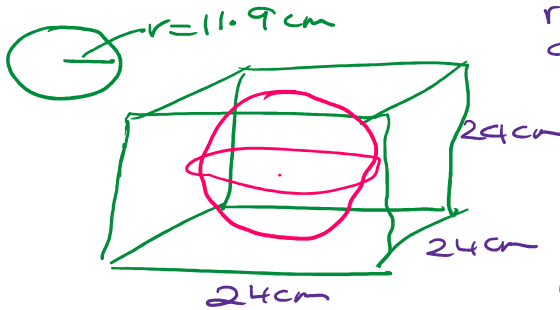
$$3 \times 2 \times 2 = 12$$

$$96 \div 12 = 8$$

$$\frac{1}{3} \times 2$$



how would you do $\frac{14}{5} = ?$
basket ball



$$\text{radius} = 11.9 \text{ cm}$$

$$\text{diameter} = 2 \times r$$

$$= 2 \times 11.9$$

so 1 basketball in box.

$$\text{Volume of box} = 24 \times 24 \times 24$$

$$= (24)^3 = 13824 \text{ cm}^3$$

volume of basketball

$$V_{\text{sphere}} = \frac{4}{3} \pi r^3$$

$$= \frac{4}{3} \times \pi \times (11.9)^3$$

$$= 7058 \text{ cm}^3$$

$$\text{empty space} = \text{box} - \text{sphere}$$

constant difference

ARITHMETIC SEQUENCE PUZZLE

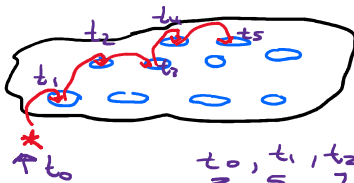
Fill in the missing numbers in the "crossword" puzzle below so that each entry is an arithmetic sequence.

arithmetic
sequences
&
geometric
sequences

?

$$\frac{46-32}{14} = 7 \text{ diff}$$

$$\text{diff} = 2$$



$$t_0, t_1, t_2, t_3$$

$$3, 5, 7, \dots$$

$$\text{diff} = +2$$

constant difference = arithmetic sequence

$$\begin{array}{c|c|c|c|c} t_0 & t_1 & t_2 & t_3 & t_4 \\ \hline 3 & 9 & 27 & 81 & 243 \end{array}$$

$$\times 3 \quad \times 3 \quad \times 3 \quad \times 3$$

constant ratio = geometric sequence

$$\frac{9}{3} = \frac{27}{9} = \frac{81}{27} \Rightarrow \text{ratio is } 3$$

Dimensions

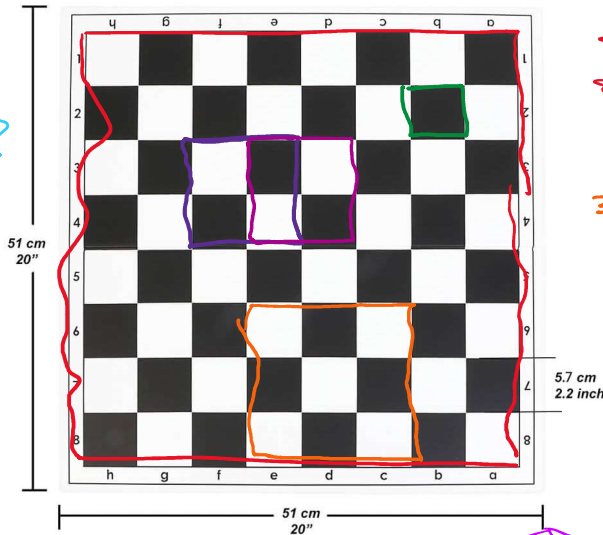


Dimensions

How many squares on a chessboard?

65??

$$64 = 8 \times 8$$



Size

8x8

1x1

2x2

3x3

how many

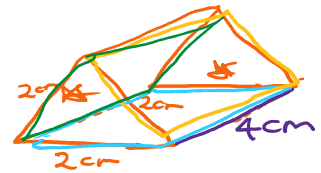
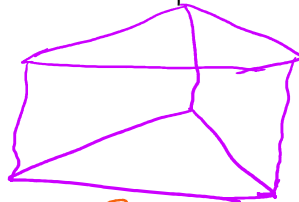
64 (8x8)

??

T.B.C

204

surface
The area of prism (3D)

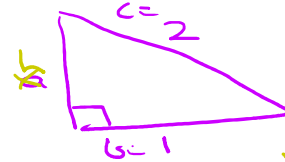


$$2 \times 8 + 2 \times 8 + 2 \times 8 + 2 \times \frac{1}{2} \times 2 \times \sqrt{3}$$

$$TSA = 24 + 2\sqrt{3}$$



$$A = \frac{1}{2} \times 2 \times \sqrt{3} = \sqrt{3}$$



$$a^2 + b^2 = c^2$$

$$a^2 = c^2 - b^2$$

$$= 2^2 - 1^2$$

$$= 4 - 1$$

$$\sqrt{a^2} = \sqrt{3}$$

$$a = \sqrt{3}$$